Time to Change Things Up: Evolving the Conversation about Results Reporting

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Reporting has been an interest of mine for about 20 years now, dating back to my first year in grad school

Salient because of a conversation at my first NCME/AERA in Montreal, April 1999

It was suggested to me that reporting was a “dead-end topic” and I might look elsewhere to contribute to our field

Clearly, I disagreed.

- Reporting *does* matter.
- Communicating test results well is not easy.
- Progress is being made, but we haven’t solved everything.

PATHWAY FOR TODAY

- Briefly trace ways in which research and practice of score reporting has progressed in recent years (and in some ways, not)
- Reflect on challenges and opportunities for reporting in 2019 and beyond
- Talk about ideas I'm exploring in my work

ART AND SCIENCE

Science: Psychometrics and statistics

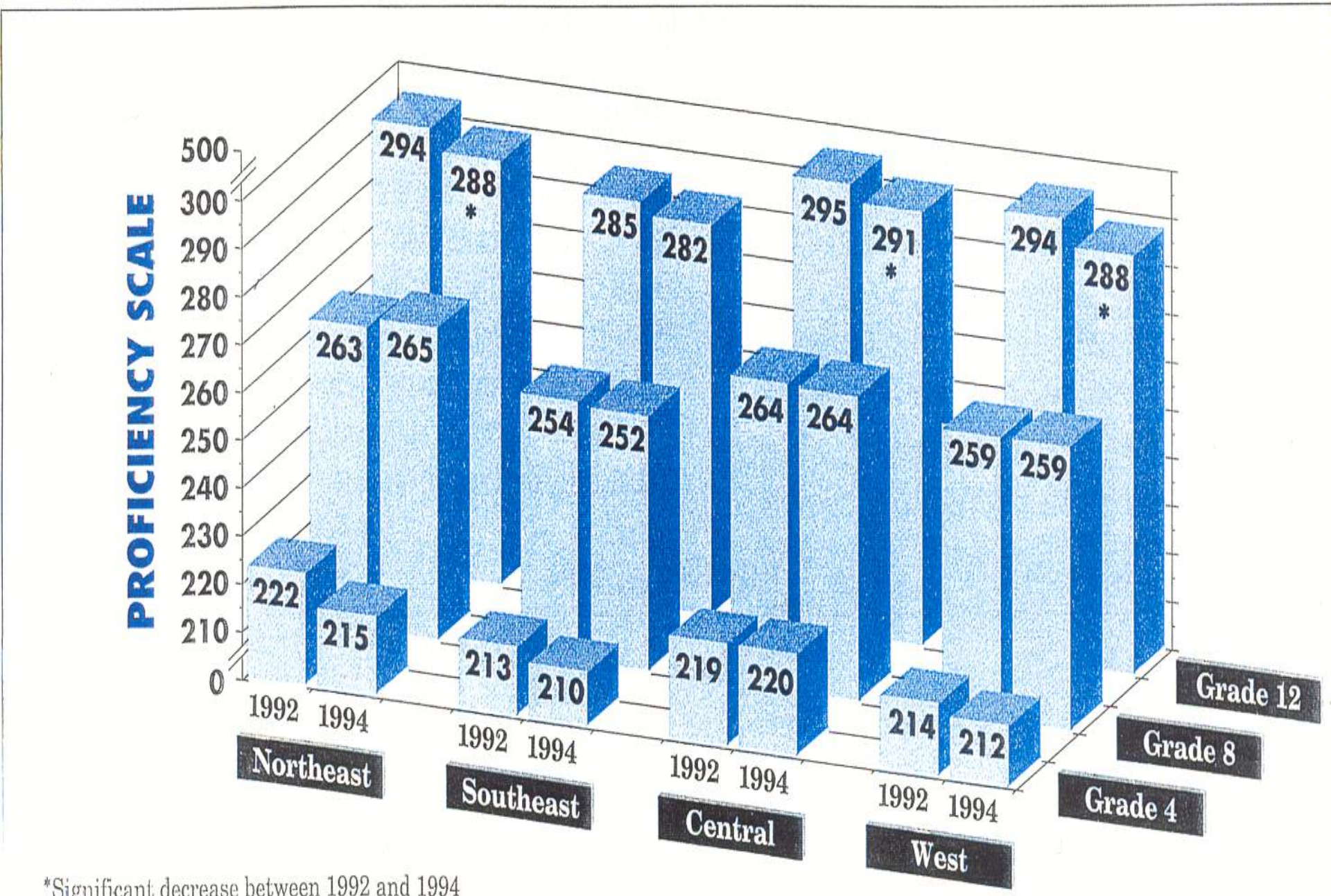
- What data should be communicated?
  - Scores, percentiles, grade equivalents, errors, written diagnostic information, skill profiles, etc.

Art: Public relations, graphic design

- What is the most effective way to communicate that data?
  - Graphs, tables, color, layout and format, etc.

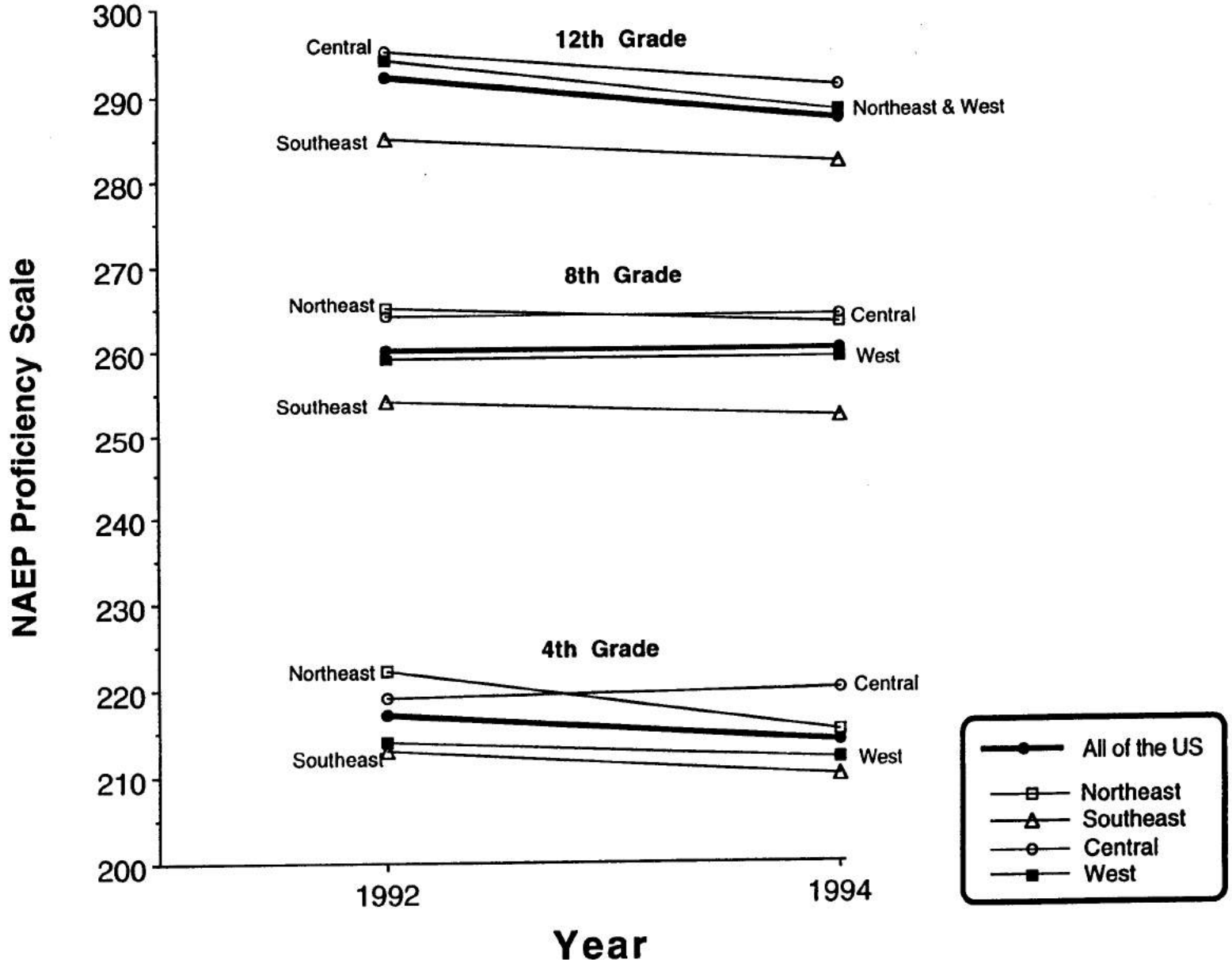
- Wainer, Hambleton, & Meara (1999) Study
  - Worked with 1994 NAEP Reading data, and carried out experimental study
  - Identified five problematic displays and redesigned them
  - Engaged 100 educators and policy makers

Figure 3. Average Reading Proficiency by Grade and by Region — NAEP 1992 and 1994



*Significant decrease between 1992 and 1994

Average Reading Proficiency by Grade and Region
NAEP 1992 and 1994



PERSPECTIVES ON REPORTING AS A NON-"DEAD END" TOPIC

- *External*: Role of tests and testing has evolved over the years, particularly in education context
  - Public perceptions, increasingly polarizing language

- *Internal*: the actions of reporting were historically viewed an add-on to the test development process
  - Often it's up to test developers, but we're perhaps not best equipped for that

BACKGROUND - INDIVIDUAL REPORTS

- Purpose: To provide information about individual performance
- Relevant stakeholders:
  - Examinees themselves
  - Families
  - Educators (teachers, profs, administrators, etc.)
  - Decisionmakers (employers, admissions, etc.)
- Relevance of granular data decreases with distance

BACKGROUND - GROUP-LEVEL REPORTS

- Purpose: To provide information about performance of groups of test-takers
- Often, groups are defined as:
  - Proximity/Geography: Class, school, district, state, etc.
  - Demographics: Race/ethnicity, sex, special needs status
- Stakeholders
  - More proximal (educators, admins., institutions)
  - Less proximal (general public, policymakers)
- Action versus information orientation shifts with distance

BACKGROUND - CONTENT ELEMENTS



REPORT PURPOSE AND/OR TEST PURPOSE



OVERALL SCORE



PERFORMANCE LEVEL (BINARY OR >2 CATEGORIES)



RELEVANT REFERENCE GROUP COMPARISON DATA



INDIVIDUAL TEST QUESTION PERFORMANCE RESULTS



SUBCONTENT AREA PERFORMANCE DETAILS



GROWTH PERCENTILES

BACKGROUND - DESIGN ELEMENTS

- Personalization
- Highlight main/key results
- Clear sectioning of report elements: Logical flow
- Mix of text, tables, and graphics
- Compartmentalized layout / Everything has a place
- Balance data/information with white space
- Minimal jargon, long technical explanations
- Color (judiciously used)
- Length: 1 vs. 2 (or more!!) pages

BACKGROUND - AUDIENCES

- Educators
- Educational Administrators
- Parents
- Students
- Politicians/Policymakers
- Media
- Business Community
- Researchers
- General Public

All of these groups have different interests and needs regarding test data and results

Likewise, they may vary in assessment needs and interests

FROM WHENCE WE CAME…

HISTORICAL CHALLENGES IN REPORTING (Or, "The World According to Ron Hambleton", when I got into this stuff)

- Scales: Reporting scales
  - Raw scores, scale scores; Percents & percentiles; IQ scores…
- Disjointed or meaningless information on some reports
- Lack of consideration of users
- Lack of error bands (and good explanations of what that means for interpretation)
- Lack of actionable diagnostic information

- Happily, things have changed greatly since then
- Reporting is indeed part of the TD conversation
  - Design processes: work by Hambleton & Zenisky; Zapata-Rivera
  - Validity: Thinking (and talking!) about actual reporting activities not at end, but *beginning*
  - Ongoing, multidisciplinary work on contents and formats
  - Engaging stakeholders in research and practice

Student Name: **Jane S. Doe**
State Student ID: **9999 123 456**
Grade: **8**

School: **Demo School**
District: **Demo District**
Test Date: **Spring 2015**

State of Washington
Office of
Superintendent of Public Instruction

OSPI

Family Report

Smarter Balanced ELA/Literacy Assessment Results

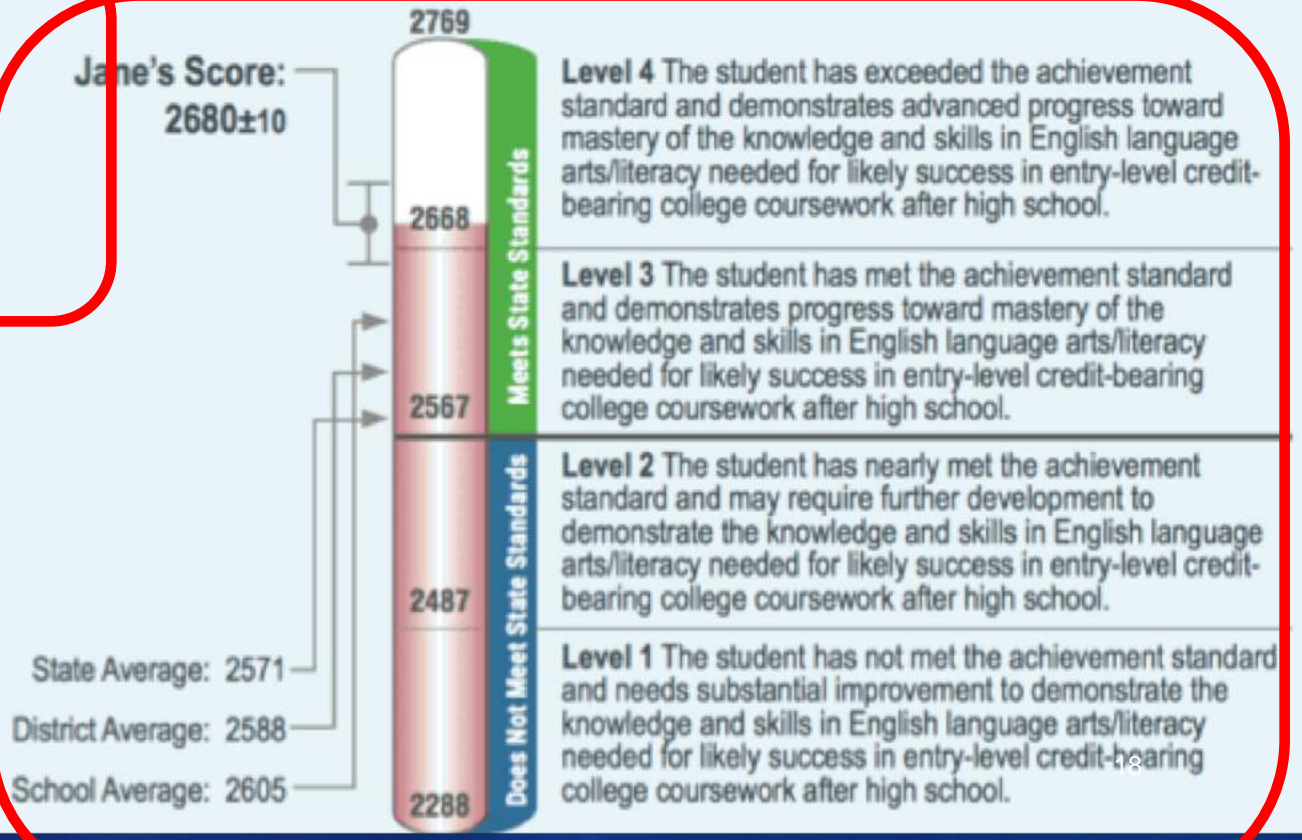
Jane's ELA/Literacy Score

2680

How does this compare?

Jane's ELA/Literacy scale score is 2680. This score is higher than the average score of eighth graders in her school, higher than that of eighth graders in her district, and higher than that of eighth graders statewide.

A student's test score can vary if the test is taken several times. If your child was tested again, it is likely that Jane would receive a score between 2670 and 2690.



Level 4 The student has exceeded the achievement standard and demonstrates advanced progress toward mastery of the knowledge and skills in English language arts/literacy needed for likely success in entry-level credit-bearing college coursework after high school.

Level 3 The student has met the achievement standard and demonstrates progress toward mastery of the knowledge and skills in English language arts/literacy needed for likely success in entry-level credit-bearing college coursework after high school.

Level 2 The student has nearly met the achievement standard and may require further development to demonstrate the knowledge and skills in English language arts/literacy needed for likely success in entry-level credit-bearing college coursework after high school.

Level 1 The student has not met the achievement standard and needs substantial improvement to demonstrate the knowledge and skills in English language arts/literacy needed for likely success in entry-level credit-bearing college coursework after high school.

USMLE United States Medical Licensing Examination®

United States Medical Licensing Examination®

Step 3 Score Report

FOR EXAMINEE USE ONLY. THIRD-PARTY USERS OF USMLE SCORES SHOULD RELY SOLELY ON OFFICIAL TRANSCRIPTS RECEIVED DIRECTLY FROM THE EXAMINEE'S USMLE REGISTRATION ENTITY.

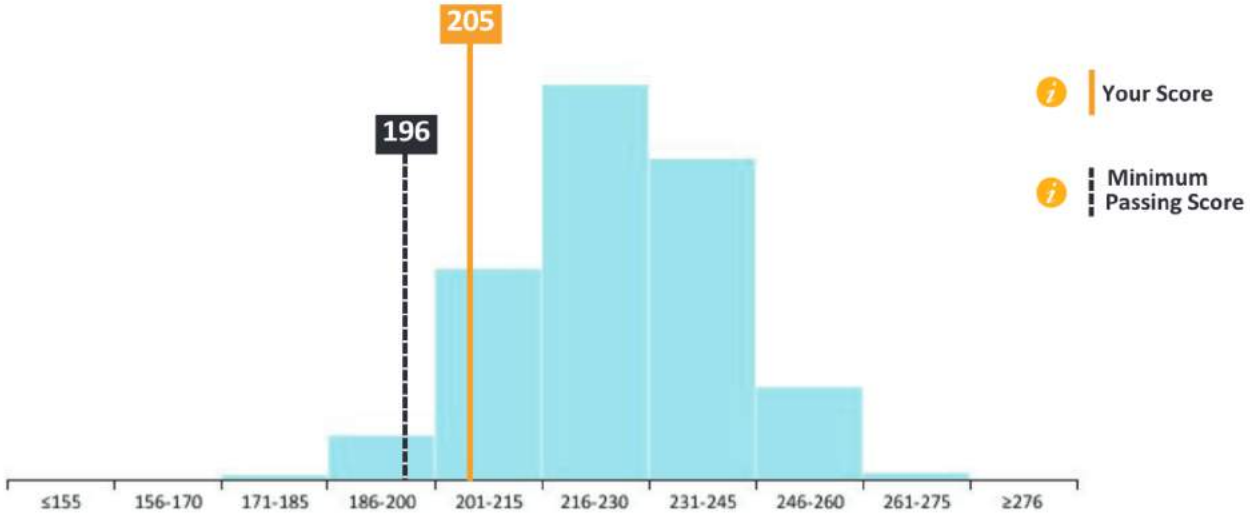
NAME: Last, First
USMLE ID: 0-000-000-0
TEST DATE: Month 1, 2018

Your Performance

Test Result	Test Score
PASS	**205**

Your Performance Compared to Other Examinees

The chart below represents the distribution of scores for recent examinees from US and Canadian medical schools taking Step 3 for the first time. Reported scores range from 1-300 with a mean of 226 and a standard deviation of 15.



If you tested repeatedly under the same conditions on a different set of items covering the same content, without learning or forgetting, your score would fall within one standard error of the estimate (SEE) of your current score two-thirds of the time. The SEE on this exam is 8 points.

Your score +/- SEE: 197 – 213

Your Evidence-Based Reading and Writing Score

380 | 160 to 760

Your Nationally Representative **18th** Sample Percentile



Your scores indicate you are close to being on track for college readiness, but you need to continue to strengthen your skills.

Let's get you back on track, so you won't have to take noncredit courses in college. You have free, personalized recommendations waiting for you on khanacademy.org/sat

Your Total Score

860 | 320 to 1520

Your Nationally Representative **28th** Sample Percentile

Keep in mind, the PSAT/NMSQT® and SAT are on the same scale.

Your score shows you how you would have scored that day on the SAT®.' How well you do depends on what you do next!

Your Math Score

480 | 160 to 760

Your Nationally Representative **47th** Sample Percentile



You are on track for college readiness.

Stay on track and continue your progress. Start now with your free, personalized recommendations waiting for you on khanacademy.org/sat

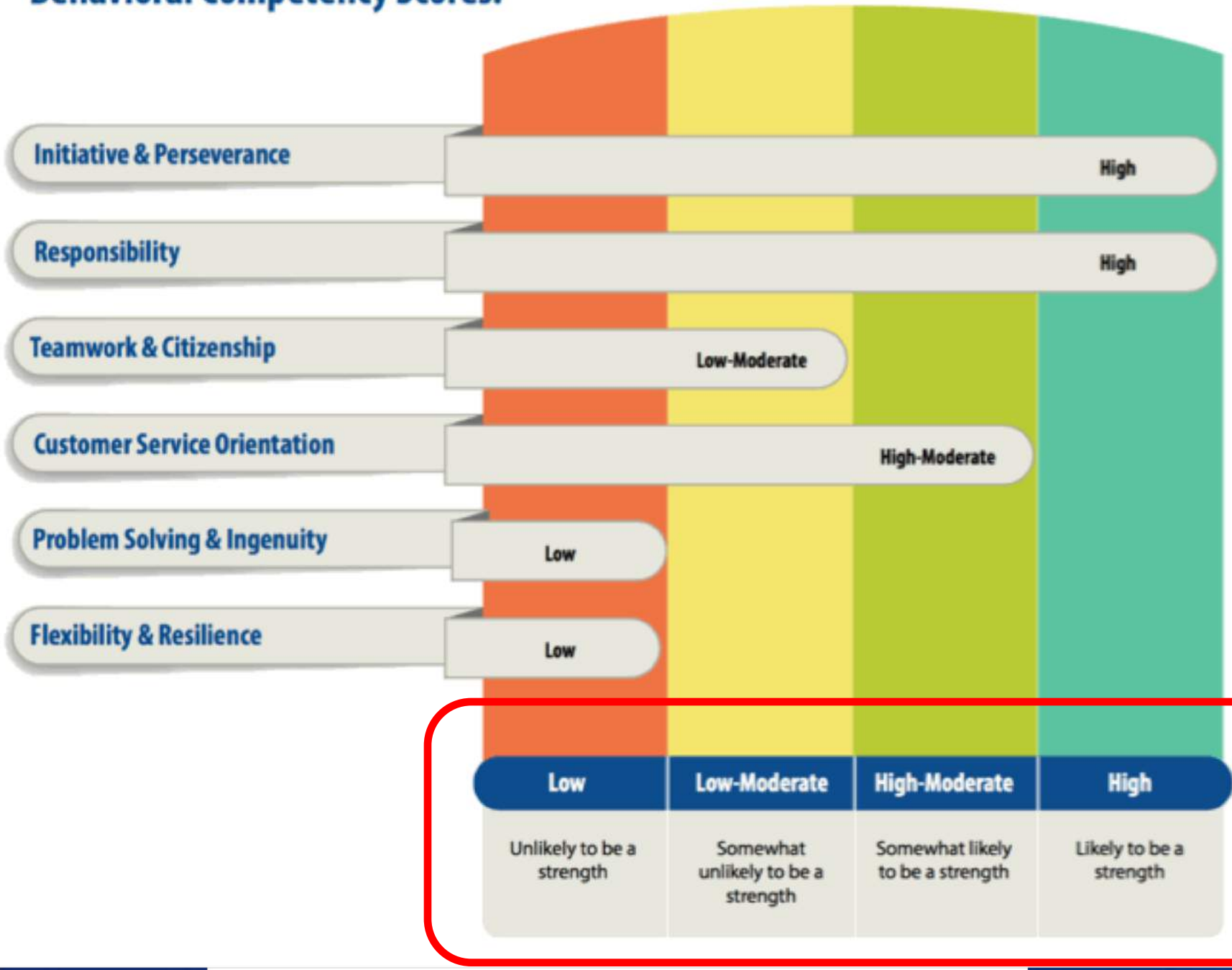
The College and Career Readiness Benchmark

Reaching your grade-level Benchmark means that you are likely on track to be ready to succeed in select first-year, credit-bearing college courses

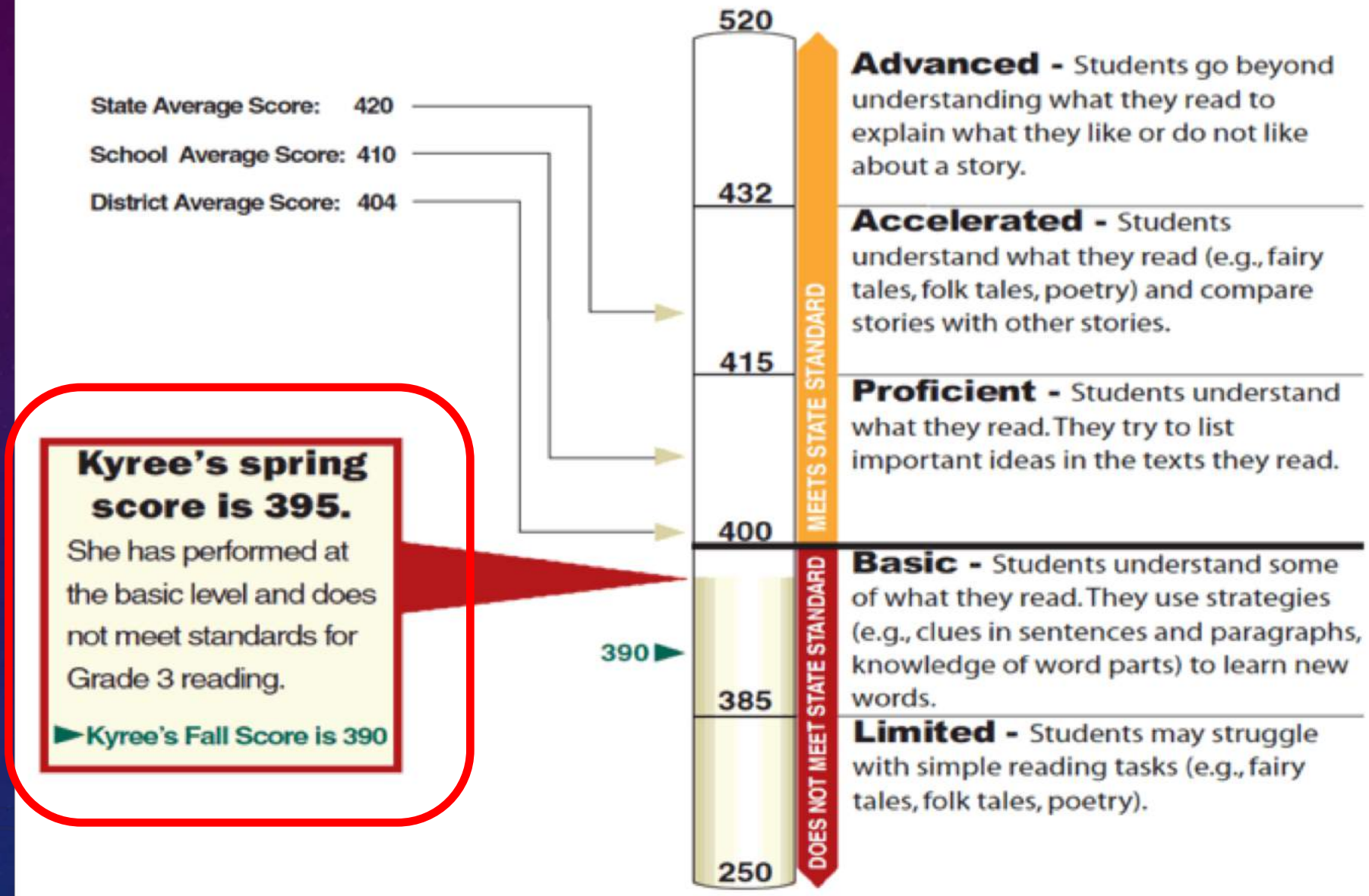
Need to strengthen skills | Approaching Benchmark | Meets or exceeds Benchmark

*The red, yellow, and green ranges in the test scores and subscores reflect your areas of strengths and weaknesses compared to the typical performance of students in your grade

Behavioral Competency Scores:



Reading Achievement Results



CONNECTICUT STATE DEPARTMENT OF EDUCATION

CSDE

Student Name: **Jacqueline Doe**
Grade: **08**
Date of Birth: **05/20/2002**
SASID: **1234567892**

School: **Demo Middle School**
District: **Demo District**
Test Date: **Spring 2015**

Overall Results

Jacqueline scored at Level 3 on the English language arts/Literacy test and scored at Level 2 on the Mathematics test.

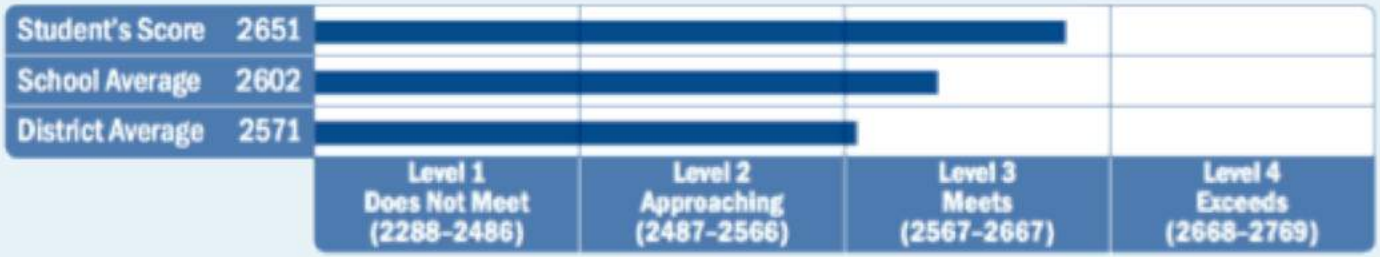
	Level 1	Level 2	Level 3	Level 4
ELA/Literacy			✓	
Mathematics		✓		

ELA/Literacy Results

Jacqueline's Total Scale Score=2651 (Score Scale Range 2288-2769)

Level 3: Meets the Achievement Level

Jacqueline has **met the achievement level** for English language arts and literacy expected for this grade. Students performing at this level are **demonstrating progress toward mastery** of English language arts and literacy knowledge and skills. Students performing at this level are on track for likely success in high school and college coursework or career training.



A student's test score can vary if the test is taken several times. If your child were tested again, it is likely that Jacqueline would receive a score between 2641 and 2661.

Areas of Knowledge and Skill	Performance
Reading	Above Standard
Writing	At/Near Standard
Listening	Above Standard
Research/Inquiry	At/Near Standard

FOR SUMMATIVE ASSESSMENT PURPOSES…

- Report development efforts are coalescing around several key elements
  - Prioritizing status as information goal - snapshot view
  - Being purposeful with space and structure
  - Using layout, font size, color, and element design to communicate key data
  - Communicating error! (Zapata-Rivera, Zwick, & colleagues)

We're good at status reporting in summative context!


BUT…
SANDS ARE SHIFTING AND AND SEISMIC CHANGES ARE AFOOT
Assessments Themselves
Information Needs and Interests
Technology and Information Graphics

1. NEXT-GEN ASSESSMENTS

- Domains are getting harder to define and/or measure
  - 21st Century Skills; Collaborative Problem Solving, etc.
  - Even traditional subjects are evolving
- Multimodal data streams and natural interfaces
  - Cross-cutting traits that play out in >1 context
- Focus on Process, not just Outcomes
  - Activity patterns within a task
- KSAs to be assessed involve interactions and dependencies
- Lack of stability at grain size of interest, often

2a. TECHNOLOGICAL UNDERPINNINGS

- Increased use of the Web for everything, including reporting
- More interactive/self-directed tools
  - Significant departure for score reporting
  - Public relations aspect of large-scale testing
  - Control of statistical ‘story’ shifts from testing agency to consumer of data
- Allow users to customize which results to display based on interests / needs
  - Blurring of lines - reporting and secondary analysis

2b. DATA VIZ AND INFORMATION GRAPHICS

- We sometimes forget why visualization works!
  - Organize data by indexing it spatially
  - Offloading cognitive processes onto automatic perceptual processes
  - Free up working memory, to help guide audiences to certain patterns / data stories

- Exploration vs. communication purposes
  - Visualizations for different goals / aims

2b. DATA VIZ AND INFORMATION GRAPHICS

- Cognitive science and related research on information processing can be quite instructive for reporting
  - Principles cited in Hegarty (2019), also edited book by McCrudden, Schraw, & Buckendahl (2015), Hattie (2009)
- Inspiration from outside of testing
  - Originated with Playfair, Tufte, etc., as cited by Wainer
  - Now: apps, activity trackers, consumer product & service companies, media…

Residential Bill Front | Residential Bill Back

ComEd
An Exelon Company

Issued 2/11/16 Account # 0000000000

SERVICE FROM 1/11/16 THROUGH 2/11/16 (31DAYS)

Residential - Single

Customer Name
Service Address
City, State, Zip
000.000.0000

Total Amount Due by 3/4/16 $71.29

Thank you for your payments totaling $77.44.

TOTAL USAGE (kWh)



Current month's reading is **actual.**

AVERAGE DAILY USE (monthly usage/days in period)

Current Month — 25° avg. temp
15.8 kWh ↑ 24% from last year

Last Month — 36° avg. temp
17.1 kWh

Last Year — 25° avg. temp
12.7 kWh

Ten 100W light bulbs for 1 hour = 1 kWh

CURRENT CHARGES SUMMARY

See reverse side for details

SUPPLY $31.98

ComEd provides your energy.

ComEd.com
1.800.334.7661



DELIVERY $30.96

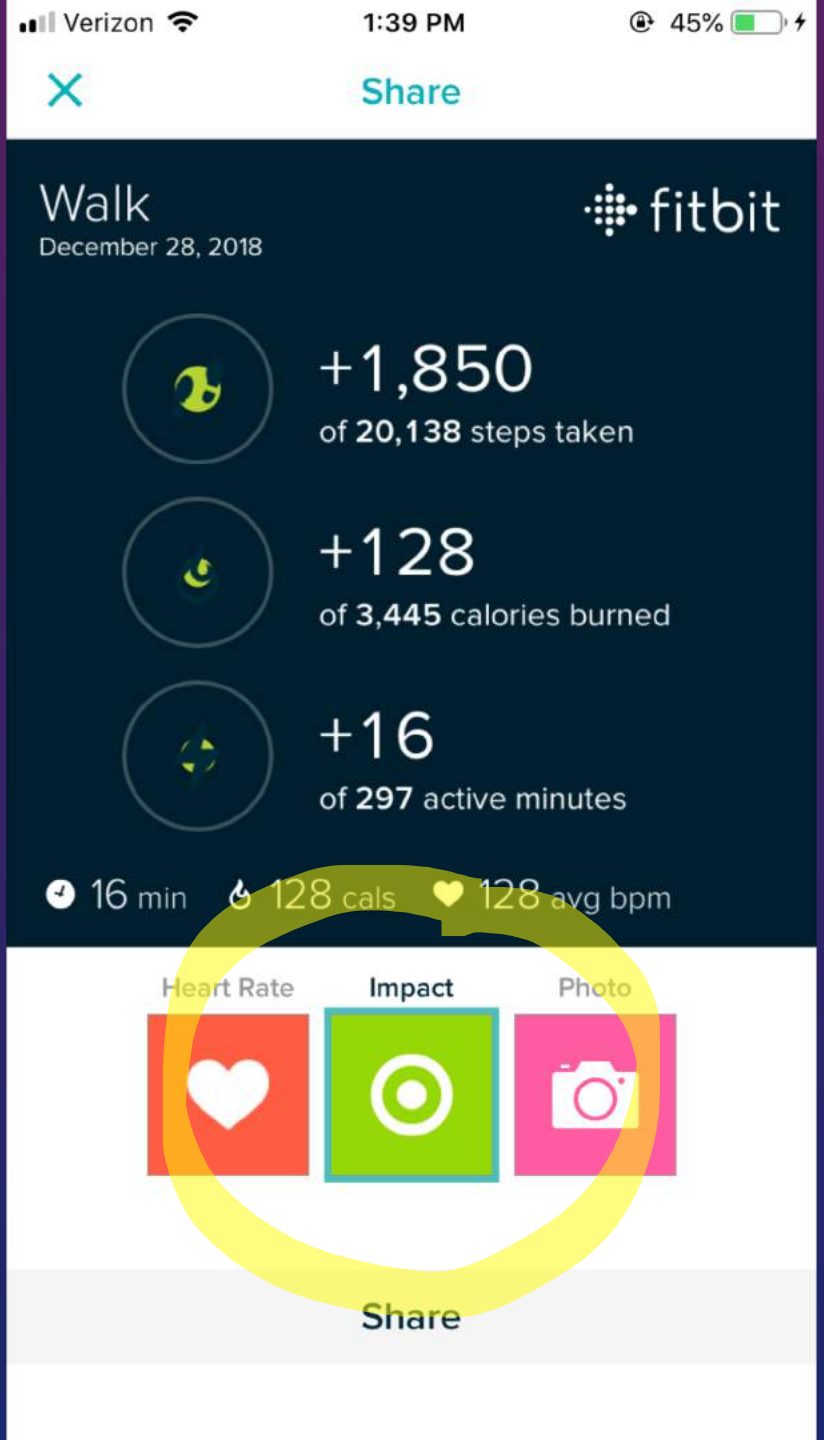
ComEd delivers electricity to your home.

ComEd.com
1.800.334.7661

TAXES & FEES $8.35

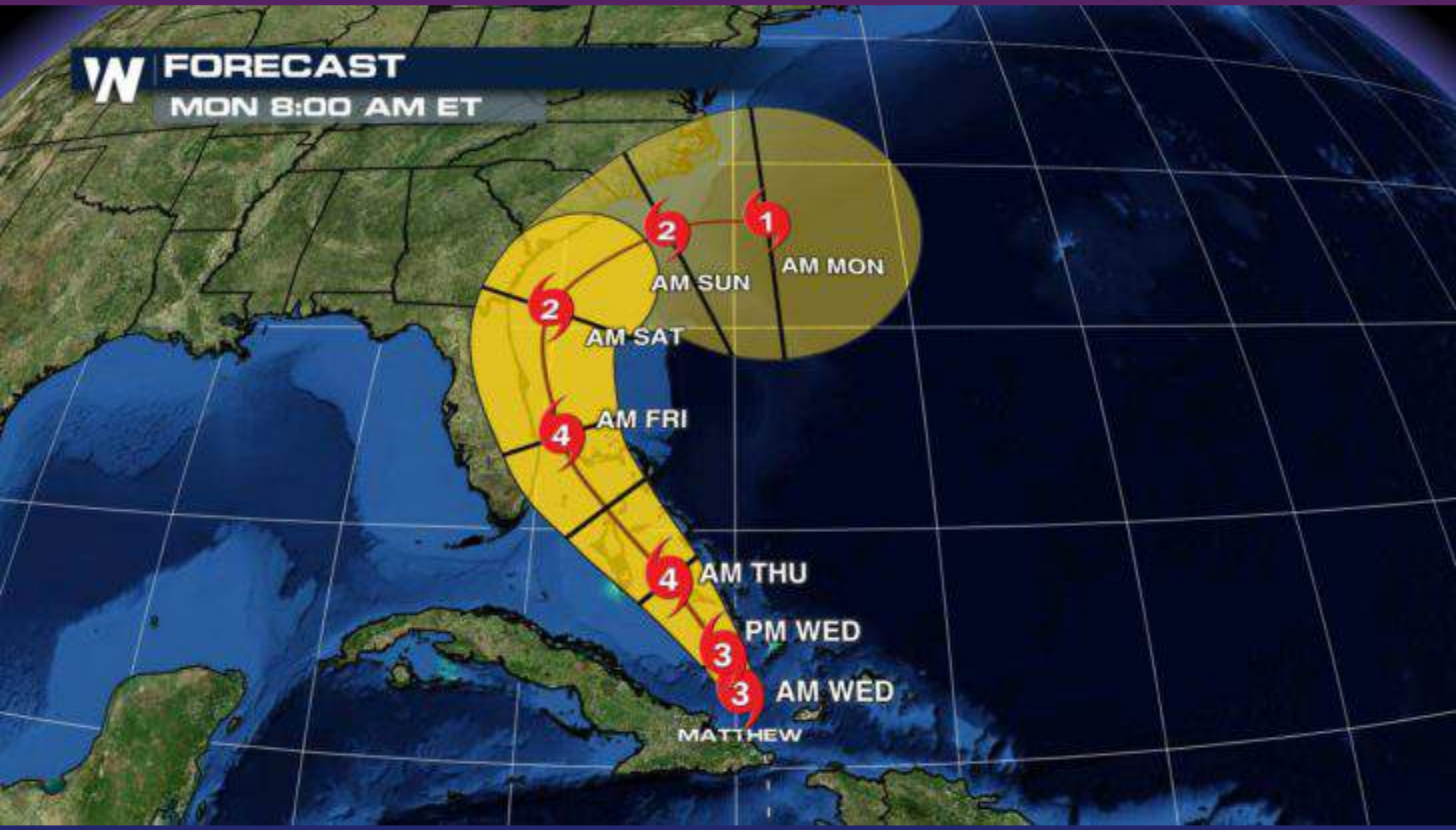
For Electric Supply Choices visit pluginillinois.org

SAMPLE

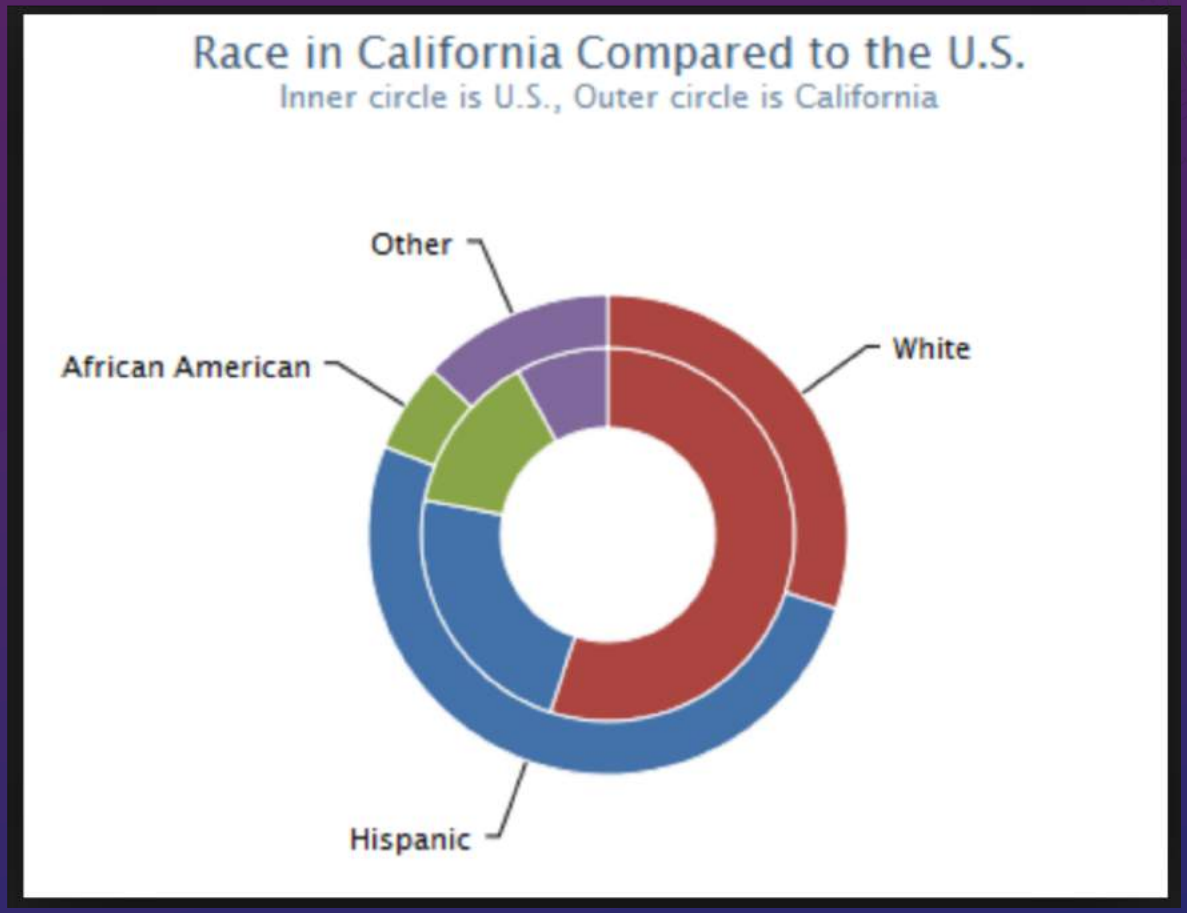


Want to change behavior?

Show impact!



Hegarty example – the width of the cone is the uncertainty at a point in time!!

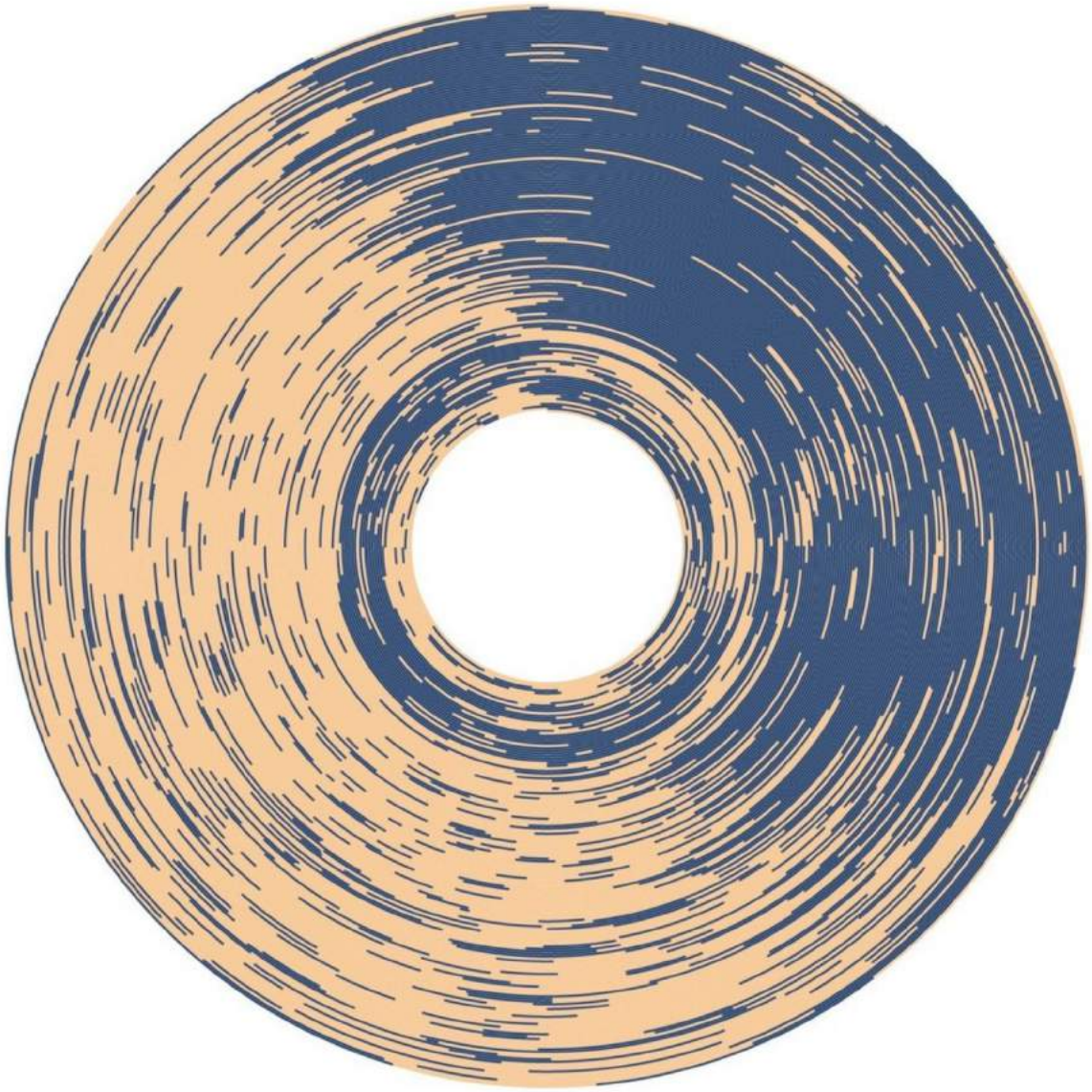


https://www.kidsdata.org/blog/?p=2811

Posted by u/andrew_elliott OC: 2 2 years ago

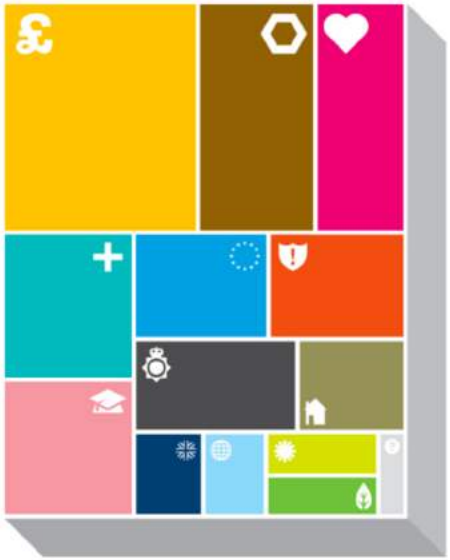
My daughters sleeping patterns for the first 4 months of her life. One continuous spiral starting on the inside when she was born, each revolution representing a single day. Midnight at the top (24 hour clock). [OC]

OC





Conservatives 29,733 words
"Forward, together"



Labour 23,818 words
"For the many not the few"



Liberal Democrats 20,739 words
"Change Britain's future"



UKIP 25,886 words
"Britain together"

Weighing their words

We compared word counts in the key parties' manifestos to work out what really mattered to them

How it works: Each manifesto is broken into coloured blocks which proportionally represent the percentage of words devoted to key topics and individual policies:

Brexit | Democracy and British values | Economy and employment | Education and childcare | Environment | Equality | Healthcare | Housing | International development and foreign aid | Ireland | Local communities | Migration | Military and defence | Miscellaneous | Multiculturalism | Online security | Policing | Scotland | The elderly and disabled

Sample policies



Conservatives

- Proposals for adult social care, later dubbed the 'dementia tax' 591 words
- Reduce net migration to tens of thousands, 345 words
- Cap on energy prices, 256 words



Labour

- Nationalise the railways, 377 words
- £30bn more for NHS, 247 words
- Scrap tuition fees, 133 words
- Increase income tax for highest earners, 108 words



Lib Dems

- Referendum on final Brexit terms, 459 words
- Legalise cannabis and end prison terms for drug possession, 187 words
- 1p extra income tax for everyone, 130 words



UKIP

- Close the Department for International Development, 525 words
- Ban the niqab/burqa, 151 words
- Not pay any Brexit 'divorce' settlement to EU, 105 words



SNP

- Keep Scotland in the EU single market, 933 words
- Second Scottish independence referendum, 254 words
- Scrap Trident, 253 words

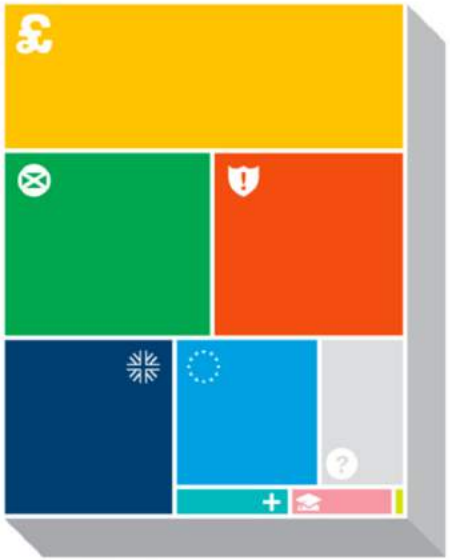


DUP

- Restore devolution and avoid direct rule 1,014 words
- 50,000 new Northern Ireland jobs, 247 words
- Reduce hospital waiting lists, 194 words



SNP 20,434 words
"Stronger for Scotland"



DUP 7,667 words
"Standing strong for Northern Ireland"

The Slow Journalism Company, click on Infographics

TURNING BACK TO TESTING…

Think Through Math

Welcome, Jonathon Ross | Logout

Home | Students | Classes | Content | Reports | Motivation | Messages 8 | Help

Reports: Individual Student Progress

View More Reports

Mrs. Smith's 8th Grade | Brittany Martin | Common Core Grade 8 | Print

Math Time



Total
09:18
Total hours:mins

School Time
Nights/Weekends

Total Performance

Total Passed: 75%



Below Grade Lessons | On Grade Lessons

Grade Level Performance

Trending at: Grade 5.6



Remediations

Placement Test Indicates Student is: Below Grade Level

20
placement test remediations
view lessons

16
failed lesson remediations
view lessons

Student Growth



3. INFORMATION INTERESTS AND NEEDS

Primary Needs

Formal test purposes and validated test uses

These are determined *a priori*, typically by the client, NOT necessarily many (most?) users

Secondary Wants

What users hope to glean from results

Status is good.
But action is where it's at (for many stakeholders)

HEIGHTENED EXPECTATIONS

- Quite simply, many users expect more.

- More:
  - Detail
  - Guidance
  - Actions
  - Help
  - Precision

More than what our current tests, *Standards* and models typically permit, at present

Your GMAT® Enhanced Score Report

This is NOT an official score report

GMAT ID: 111111111111 | VIENNA, UNITED STATES | JANUARY 20, 2015 | APPOINTMENT ID: 777777777

PERCENTILE RANKING 0 20 40 60 80 100

PERCENTILE RANKINGS COMPARE YOUR PERFORMANCE TO 3 YEARS OF GMAT EXAMS
SAMPLE SIZE: 782,458

TIME MANAGEMENT BY SECTION



MEAN TIME

MEAN TIME FOR ALL TEST TAKERS:

1.87 minutes

Definition

For all GMAT Exam scores recorded in the past three years the mean time to complete a question in the Integrated Reasoning, Verbal, and Quantitative sections is 1.87 minutes. Use this number to evaluate how well you managed your time.

SUMMARY

- Your GMAT Total score of 640 is higher than 72% of GMAT Exam scores recorded in the past three years.
  - Your Integrated Reasoning score of 3 is higher than 25% of GMAT Exam scores recorded in the past three years.
  - Your Verbal score of 34 is higher than 71% of GMAT Exam scores recorded in the past three years.
  - Your Quantitative score of 44 is higher than 58% of GMAT Exam scores recorded in the past three years.
- Your Quantitative score of 44 and your Verbal score of 34 were used to calculate your GMAT Total score. Especially strong or weak scores on either section significantly impacts the GMAT Total score.
- Time management is an important factor in the GMAT Exam scoring. Failing to complete all questions can impact your final score.

THE WICKEDEST REPORTING CHALLENGE

- The difficult, murky, thorny, troublesome, gray area of subscores, subdomains, and next steps
  - Great research by Sinharay, Haberman, et al.
- Our Unreliability Problem
  - Typically, relatively few items from each KSA
  - How reliable are granular results?
  - How useful are these results?

- Tension of “right proper” psychometrics and user perceptions of actionable results at useful grain size

Your child's performance will also be broken down into subcategories within each subject. In each subcategory, your child will be provided a rating of one to three stars, with a description of your child's performance in this specific area and the support needed moving forward.

READING PERFORMANCE

★★☆ MODERATE PERFORMANCE

STATE PERCENT OF STUDENTS AT EACH RATING

★★★ STRONG PERFORMANCE	★★☆ MODERATE PERFORMANCE	★☆☆ WEAK PERFORMANCE
30%	50%	20%

LITERARY TEXT

★★★ STRONG PERFORMANCE

In this area, your student is able to read and analyze grade-appropriate fiction, drama, and poetry very well and is prepared for further studies.

INFORMATIONAL TEXT

★★☆ MODERATE PERFORMANCE

Your student can read and analyze grade-appropriate non-fiction, including texts about history, science, art, and music. Your student may need additional support to be fully prepared for further studies.

VOCABULARY

★☆☆ WEAK PERFORMANCE

Your student will need significant support in using context to determine what words and phrases mean in grade-appropriate texts.

WRITING PERFORMANCE

★☆☆ WEAK PERFORMANCE

STATE PERCENT OF STUDENTS AT EACH RATING

★★★ STRONG PERFORMANCE	★★☆ MODERATE PERFORMANCE	★☆☆ WEAK PERFORMANCE
5%	45%	50%

WRITTEN EXPRESSION

★★☆ MODERATE PERFORMANCE

Your student can compose well-developed, organized, and clear writing, using details from what he/she read, but may need additional support to be fully prepared for further studies.

KNOWLEDGE & USE OF LANGUAGE CONVENTIONS

★★★ STRONG PERFORMANCE

Your student can compose writing using the rules of standard English (including those for grammar, spelling, and usage) and is prepared for further studies.

LEGEND

★★★ STRONG PERFORMANCE	★★☆ MODERATE PERFORMANCE	★☆☆ WEAK PERFORMANCE
Prepared for further studies	May need additional support to be fully prepared for further studies	Will need significant support for further studies

READING

Reading score range: 10 to 90	Average of students just meeting expectations	School average
Your child's score	50	52
44	District average 48	State average 45

LITERARY TEXT

In this area, your child did not do as well as students who met the expectations.

Students meet expectations by showing they can read and analyze grade appropriate fiction, drama and poetry.

INFORMATIONAL TEXT

In this area, your child did almost as well as students who met the expectations.

Students meet expectations by showing they can read and analyze grade-appropriate non-fiction, including texts about history, science, art, and music.

VOCABULARY

In this area, your child did as well as or better than students who met the expectations.

Students meet expectations by showing they can use context to determine what words and phrases mean in grade-appropriate texts.

WRITING

Writing score range: 10 to 60	Average of students just meeting expectations	School average
Your child's score:	35	39
30	District average 35	State average 31

WRITING EXPRESSION

In this area, your child did not do as well as students who met the expectations.

Students meet expectations by showing they can compose well-developed, organized, and clear writing, using details from what they have read.

KNOWLEDGE AND USE OF LANGUAGE CONVENTIONS

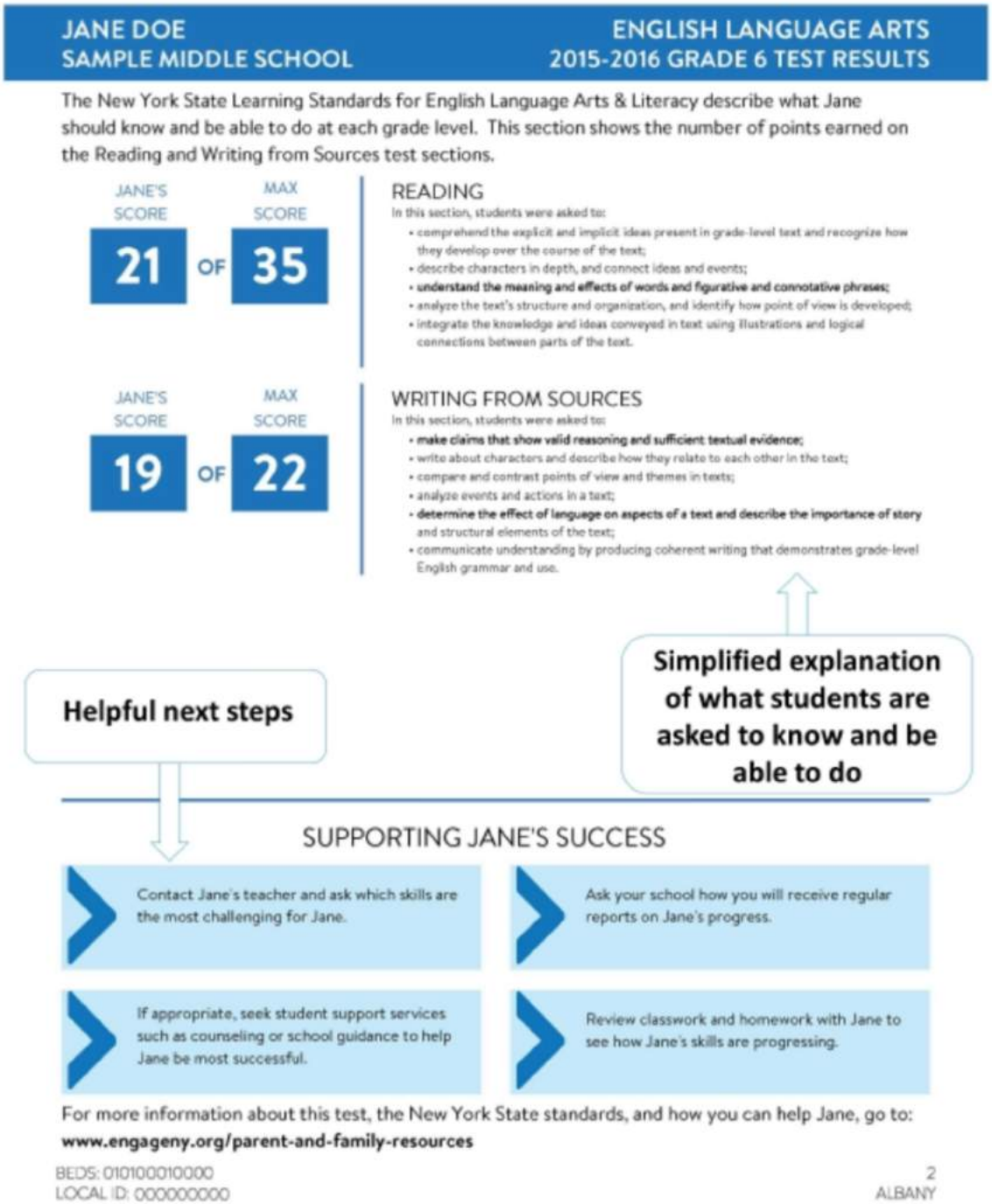
In this area, your child did as well as or better than students who met the expectations.

Students meet expectations by showing they can compose writing using the rules of standard English, including those for grammar, spelling, and usage.

LEGEND

Below Expectations | Nearly Meets Expectations | Meets or Exceeds Expectations

To see selected questions from the test, visit understandthescore.org.



JANE DOE
SAMPLE MIDDLE SCHOOL

ENGLISH LANGUAGE ARTS
2015-2016 GRADE 6 TEST RESULTS

The New York State Learning Standards for English Language Arts & Literacy describe what Jane should know and be able to do at each grade level. This section shows the number of points earned on the Reading and Writing from Sources test sections.

JANE'S SCORE: 21 OF MAX SCORE: 35

READING

In this section, students were asked to:

- comprehend the explicit and implicit ideas present in grade-level text and recognize how they develop over the course of the text;
- describe characters in depth, and connect ideas and events;
- understand the meaning and effects of words and figurative and connotative phrases;
- analyze the text's structure and organization, and identify how point of view is developed;
- integrate the knowledge and ideas conveyed in text using illustrations and logical connections between parts of the text.

JANE'S SCORE: 19 OF MAX SCORE: 22

WRITING FROM SOURCES

In this section, students were asked to:

- make claims that show valid reasoning and sufficient textual evidence;
- write about characters and describe how they relate to each other in the text;
- compare and contrast points of view and themes in texts;
- analyze events and actions in a text;
- determine the effect of language on aspects of a text and describe the importance of story and structural elements of the text;
- communicate understanding by producing coherent writing that demonstrates grade-level English grammar and use.

Simplified explanation of what students are asked to know and be able to do

Helpful next steps

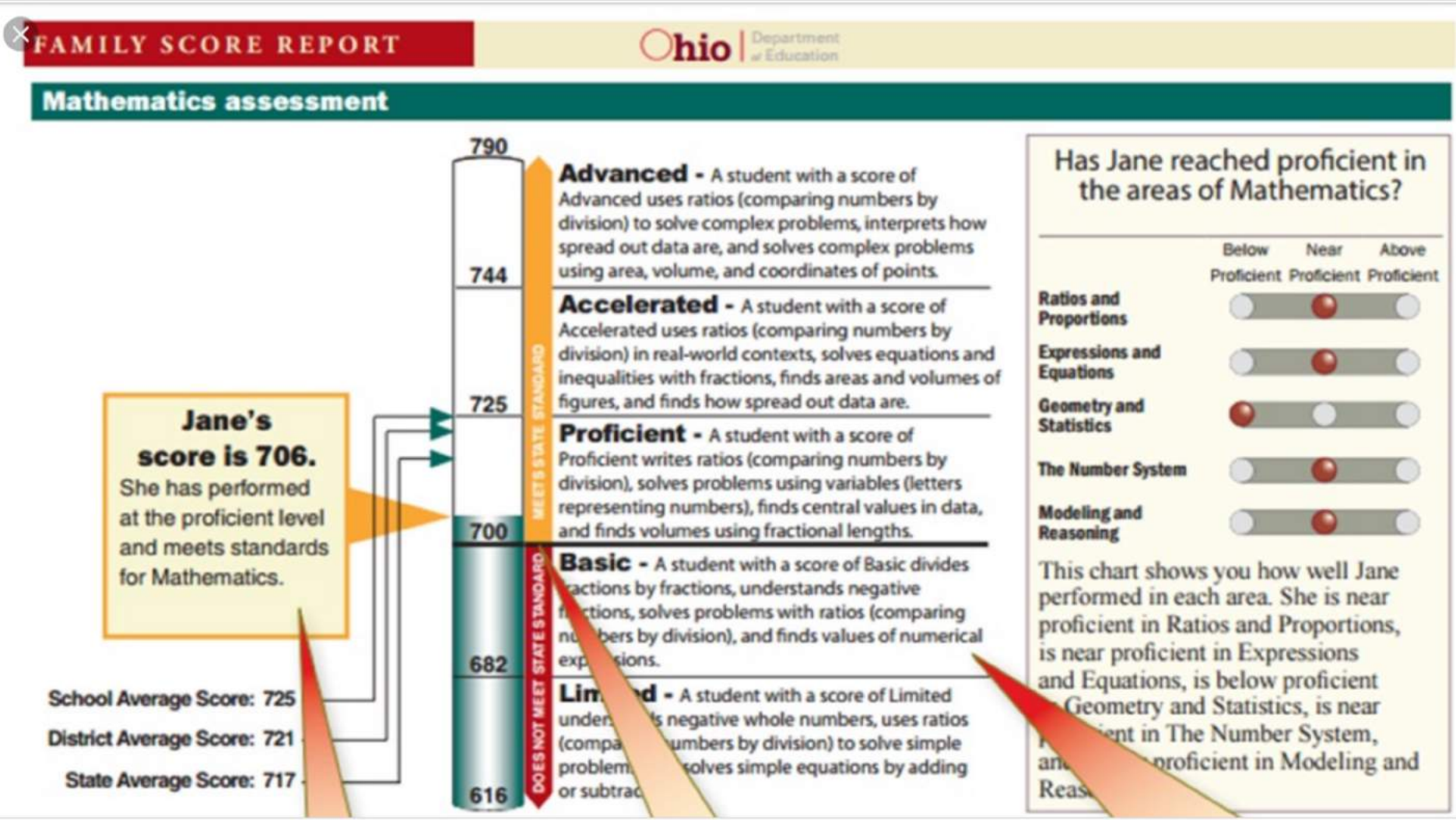
SUPPORTING JANE'S SUCCESS

- Contact Jane's teacher and ask which skills are the most challenging for Jane.
- Ask your school how you will receive regular reports on Jane's progress.
- If appropriate, seek student support services such as counseling or school guidance to help Jane be most successful.
- Review classwork and homework with Jane to see how Jane's skills are progressing.

For more information about this test, the New York State standards, and how you can help Jane, go to:
www.engageny.org/parent-and-family-resources

BEDS: 010100010000
LOCAL ID: 000000000

2
ALBANY

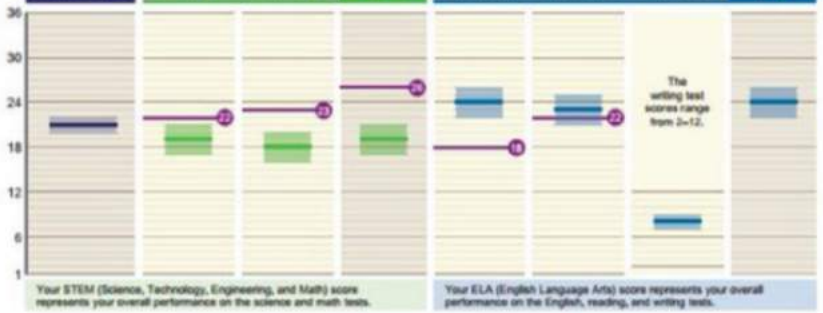

FAMILY SCORE REPORT
Ohio | Department of Education
Mathematics assessment
790
744
725
700
682
616
MEETS STATE STANDARD
DOES NOT MEET STATE STANDARD
Advanced - A student with a score of Advanced uses ratios (comparing numbers by division) to solve complex problems, interprets how spread out data are, and solves complex problems using area, volume, and coordinates of points.
Accelerated - A student with a score of Accelerated uses ratios (comparing numbers by division) in real-world contexts, solves equations and inequalities with fractions, finds areas and volumes of figures, and finds how spread out data are.
Proficient - A student with a score of Proficient writes ratios (comparing numbers by division), solves problems using variables (letters representing numbers), finds central values in data, and finds volumes using fractional lengths.
Basic - A student with a score of Basic divides fractions by fractions, understands negative numbers, solves problems with ratios (comparing numbers by division), and finds values of numerical expressions.
Limited - A student with a score of Limited understands negative whole numbers, uses ratios (comparing numbers by division) to solve simple problems, solves simple equations by adding or subtracting
Jane's score is 706.
She has performed at the proficient level and meets standards for Mathematics.
School Average Score: 725
District Average Score: 721
State Average Score: 717
Has Jane reached proficient in the areas of Mathematics?
Below Proficient
Near Proficient
Above Proficient
Ratios and Proportions
Expressions and Equations
Geometry and Statistics
The Number System
Modeling and Reasoning
This chart shows you how well Jane performed in each area. She is near proficient in Ratios and Proportions, is near proficient in Expressions and Equations, is below proficient in Geometry and Statistics, is near proficient in The Number System, and is near proficient in Modeling and Reasoning.

ANN C TAYLOR (ACT ID: -54116290)
WHEAT RIDGE SR HIGH SCHOOL (061-450)
TEST DATE: APRIL 2017 | NATIONAL

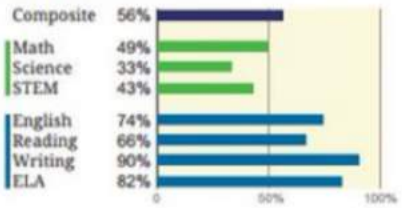
The ACT

Student Report

21 COMPOSITE	19 MATH	18 SCIENCE	19 STEM	24 ENGLISH	23 READING	08 WRITING	24 ELA



US Rank



State Rank



Your Score
Your Score
Your Score Range

ACT College Readiness Benchmarks
Readiness Benchmark
If your score is at or above the Benchmark, you have at least a 50% chance of obtaining a B or higher or about a 75% chance of obtaining a C or higher in specific first-year college courses in the corresponding subject area. There are currently no Benchmarks for ELA or writing.

Your Score Range
Test scores are estimates of your educational development. Think of your true achievement on this test as being within a range that extends about one standard error of measurement, or about 1 point for the Composite and writing scores, and 2 points for STEM, ELA, and the other test scores, above and below your score.

US & State Rank
Your ranks tell you the approximate percentages of recent high school graduates in the US and your state who took the ACT® test and received scores that are the same as or lower than your scores. For example, a rank of 55 for your Composite score means 55% of students earned that Composite score or below.

Detailed Results

MATH 19

			ACT Readiness Range
Preparing for Higher Math	22 of 35	63%	
• Number & Quantity	5 of 5	100%	
• Algebra	5 of 8	63%	
• Functions	6 of 8	75%	
• Geometry	4 of 8	50%	
• Statistics & Probability	2 of 6	33%	
Integrating Essential Skills	11 of 25	44%	
Modeling	9 of 22	41%	

SCIENCE 18

Interpretation of Data	9 of 16	56%
Scientific Investigation	7 of 10	70%
Evaluation of Models, Inferences & Experimental Results	5 of 14	36%

ACT Composite Score: ACT math, science, English, and reading test scores and the Composite score range from 1 to 36. For each test, we converted your number of correct answers into a score within that range. Your Composite score is the average of your scores on the four subjects rounded to the nearest whole number. If you left any test completely blank, that score is reported as two dashes and no Composite score is computed.

ACT Readiness Range: This range shows where a student who has met the ACT College Readiness Benchmark on this assessment would typically perform.

ENGLISH 24

			ACT Readiness Range
Production of Writing	16 of 23	70%	
Knowledge of Language	8 of 12	67%	
Conventions of Standard English	29 of 40	73%	

READING 23

Key Ideas & Details	18 of 24	75%	
Craft & Structure	6 of 11	55%	
Integration of Knowledge & Ideas	3 of 5	60%	
Understanding Complex Texts			Below / Proficient / Above

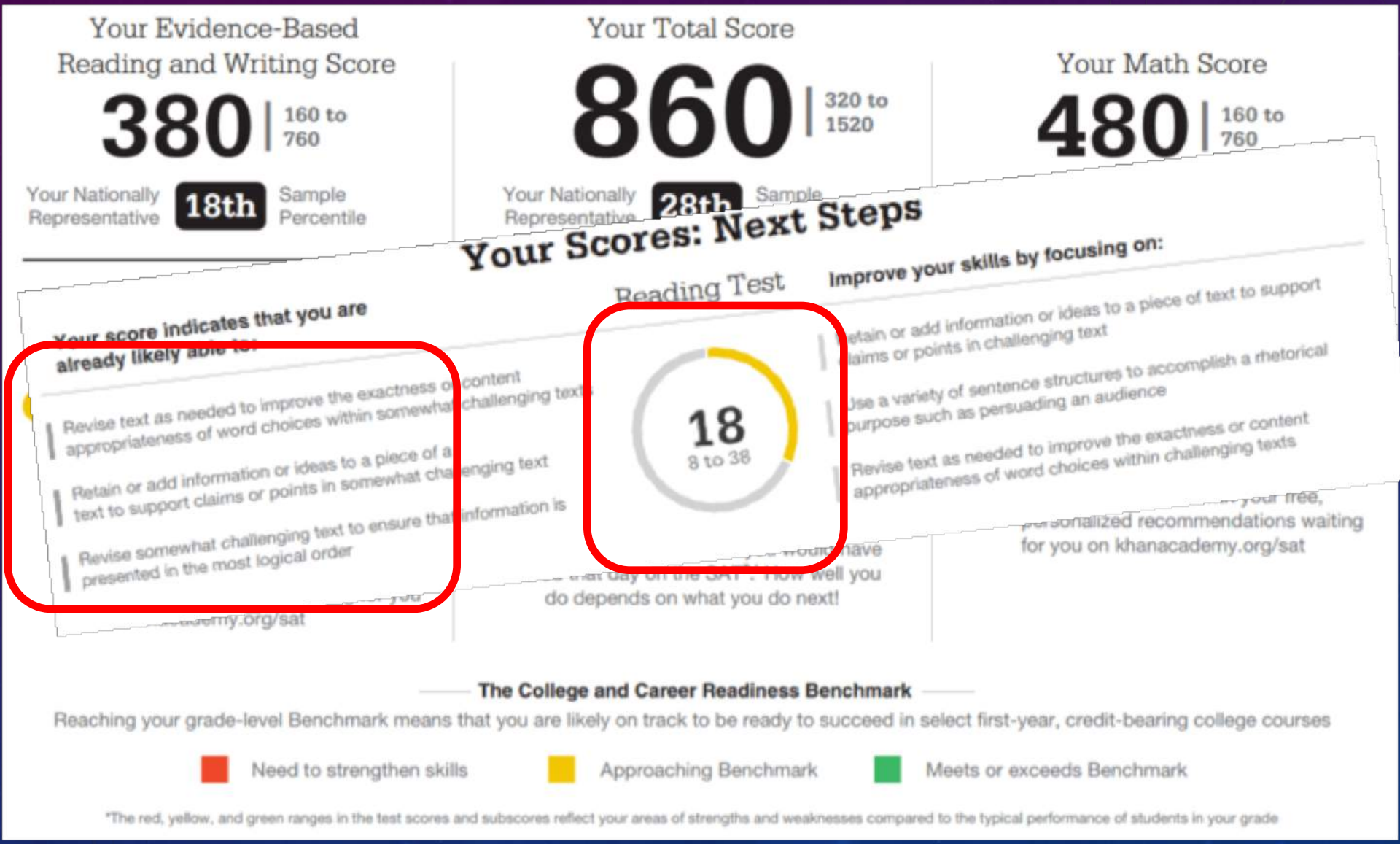
Understanding Complex Texts: This indicator lets you know if you are understanding the central meaning of complex texts at a level that is needed to succeed in college courses with higher reading demand.

WRITING 08

Ideas & Analysis	8
Development & Support	8
Organization	9
Language Use & Conventions	8

If you took the writing test, your essay was scored on a scale of 1 to 6 by two raters in each of the four writing domains. These domains represent essential skills and abilities that are necessary to meet the writing demands of college and career. Your domain scores, ranging from 2 to 12, are a sum of the two raters' scores. Your writing score is the average of your four domain scores rounded to the nearest whole number. To learn more about your writing score, visit **www.act.org/the-act/writing-scores.**

Dashes (--) indicate information was not provided or could not be calculated.


Your Evidence-Based Reading and Writing Score
380
160 to 760
Your Nationally Representative
18th
Sample Percentile
Your Total Score
860
320 to 1520
Your Nationally Representative
28th
Sample
Your Math Score
480
160 to 760
Your Scores: Next Steps
Reading Test
18
8 to 38
Improve your skills by focusing on:
Revise text as needed to improve the exactness of word choices within somewhat challenging text
Retain or add information or ideas to a piece of a text to support claims or points in somewhat challenging text
Revise somewhat challenging text to ensure that information is presented in the most logical order
Use a variety of sentence structures to accomplish a rhetorical purpose such as persuading an audience
Revise text as needed to improve the exactness or content appropriateness of word choices within challenging texts
do depends on what you do next!
for you on khanacademy.org/sat
The College and Career Readiness Benchmark
Reaching your grade-level Benchmark means that you are likely on track to be ready to succeed in select first-year, credit-bearing college courses
Need to strengthen skills
Approaching Benchmark
Meets or exceeds Benchmark
*The red, yellow, and green ranges in the test scores and subscores reflect your areas of strengths and weaknesses compared to the typical performance of students in your grade

FAQs

What is the Smarter Balanced ELA/ Literacy Assessment?

This assessment is aligned to new academic standards for ELA/literacy. These standards identify what students should know and be able to do to graduate high school ready for college and the workplace. They challenge students to develop a deeper understanding of subject matter, learn how to think critically, and apply what they are learning to the real world.

What is a claim?

ELA claims are broad statements that identify the knowledge and skills students should know and be able to do as they progress toward college and career readiness. How your child performed on each claim is found in the columns to the right.

How did Jane perform on the different claims of the test?

Claim 1: Reading	Claim 2: Listening and Speaking	Claim 3: Writing	Claim 4: Research/Inquiry
Below Standard	Above Standard	Above Standard	At/Near Standard
Student has difficulty reading closely and analytically to comprehend a range of increasingly complex literary and informational texts.	Student can employ effective speaking and listening skills for a range of purposes and audiences.	Student can produce effective and well-grounded writing for a range of purposes and audiences.	Student may be able to engage in research and inquiry to investigate topics, and to analyze, integrate, and present information.

TO RECAP THE BIG CHANGES:



Tests are changing



Technology and data visualizations are evolving



Data ‘wants’ are growing

… so how do we move forward?

CHANGING THINGS UP, #1

- Let's think about how we think about data and reports
  - What we're good at is only part of what some stakeholders want
  - Pursuit of psychometric perfection is paralyzing progress

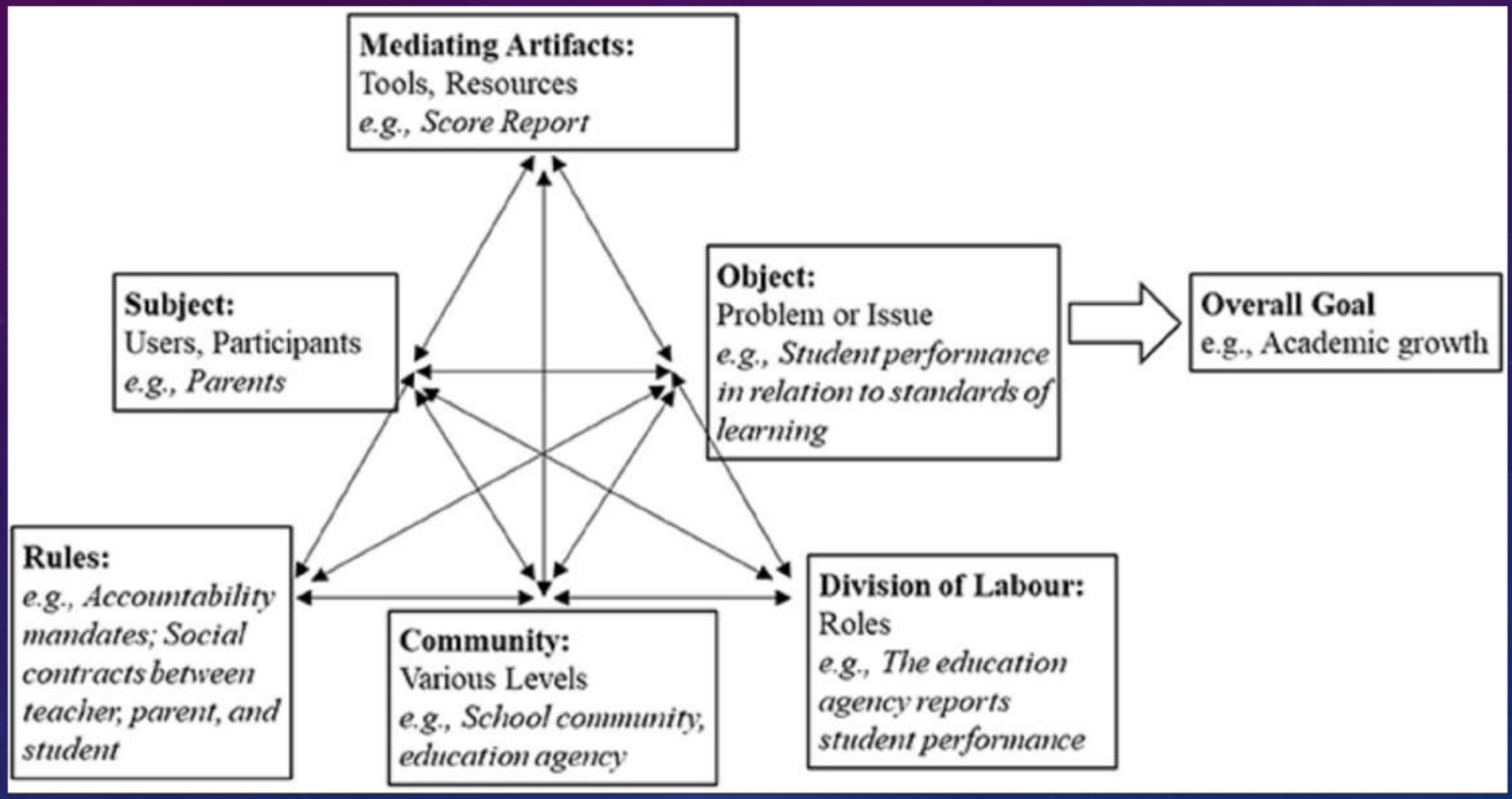
- What balance can we strike?
- For lower-stakes uses, can we think about being more forthcoming about data, acknowledging limitations?

CHANGING THINGS UP, #2

- New work by Roduta Roberts, Gotch, & Lester (2018)
  - Focus on Language of reporting
  - Discourse analysis of SR language, to ID patterns

- Idea of *activity theory*
  - "analysis using this framework provides a window into how different individuals (i.e., subjects) who are all working toward the same overall goal within a system may have different priorities and motives, perhaps acting on different objects within the system."

CHANGING THINGS UP, #2



- Displays of Information, Knowledge Claims, and Doing Accountability

CHANGING THINGS UP, #2

- Maybe activity theory is ‘right’, maybe it’s not
- But it’s a different approach to developing and evaluating reports
- New ideas, different perspectives on how we talk about test results might be helpful to move us past what we’re good at to what we need to work on as a field
  - especially with new & emerging tests, contexts, constructs

CHANGING THINGS UP, #3

- Evolution in how people interact with information
  - Explanatory and Instructive
  - Exploratory and Constructive

- As the agents of reporting, these orientations inform the choices we make in developing and evaluating reporting systems

- Not just a job for psychometricians
- Please: Assemble a team (Slater, Livingston, Lester, 2019)

CHANGING THINGS UP, #4

- Figure out our angles for different stakeholder groups
  - Adopt a question-based approach (Van Winkle, Vezzu, & Zapata-Rivera, 2011)
- Build interactive report around specific question(s)
  - Graphical representation of answer
  - Results description
  - Highlights of results
  - Limitations
  - Statistical caveats

SO, CONCLUDING QUESTIONS…

- Focus of many conversations about score reporting is how to develop a report
  - What do we put on there?
  - What color for the bars?
- Great, *but*…
- If goal really is promoting meaning:
  - Let's take a look, and have some hard conversations about what we are reporting, to whom?

QUESTIONING OUR STATUS QUO

- Will our current reporting conventions and approaches work with next-gen assessments? How well?
- Can we begin to harness how people consume and use information now and in the coming years?
- How do we reconcile have-to's and want-to's to make information available and useful at different levels of stakes?

Conceptually return to roots of report development

- Reporting responsibility is more than just the document with performance data

In reporting systems:

- Acknowledge range of uses, develop flexibilities in our approaches
- Allocate real resources to develop interpretive resources
- Be creative about formats and access

Thanks!

azenisky@educ.umass.edu